Advanced Formal Tools

CTL Modle Checker

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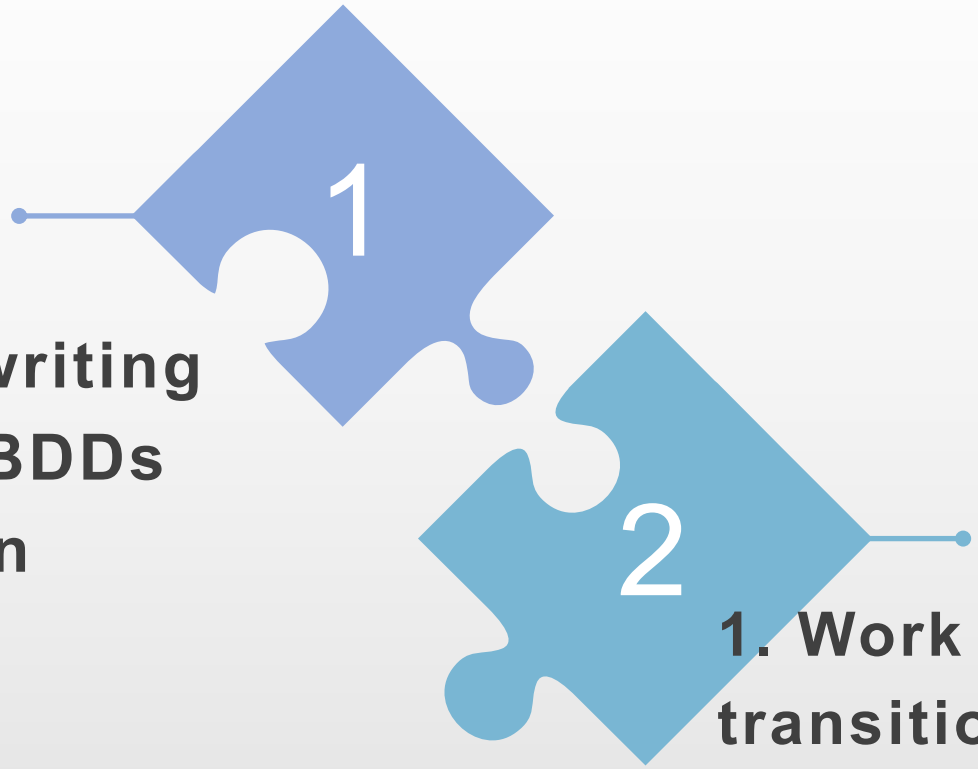
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Current Status

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2.Understandingof BDDs
3.Partition transition relation3

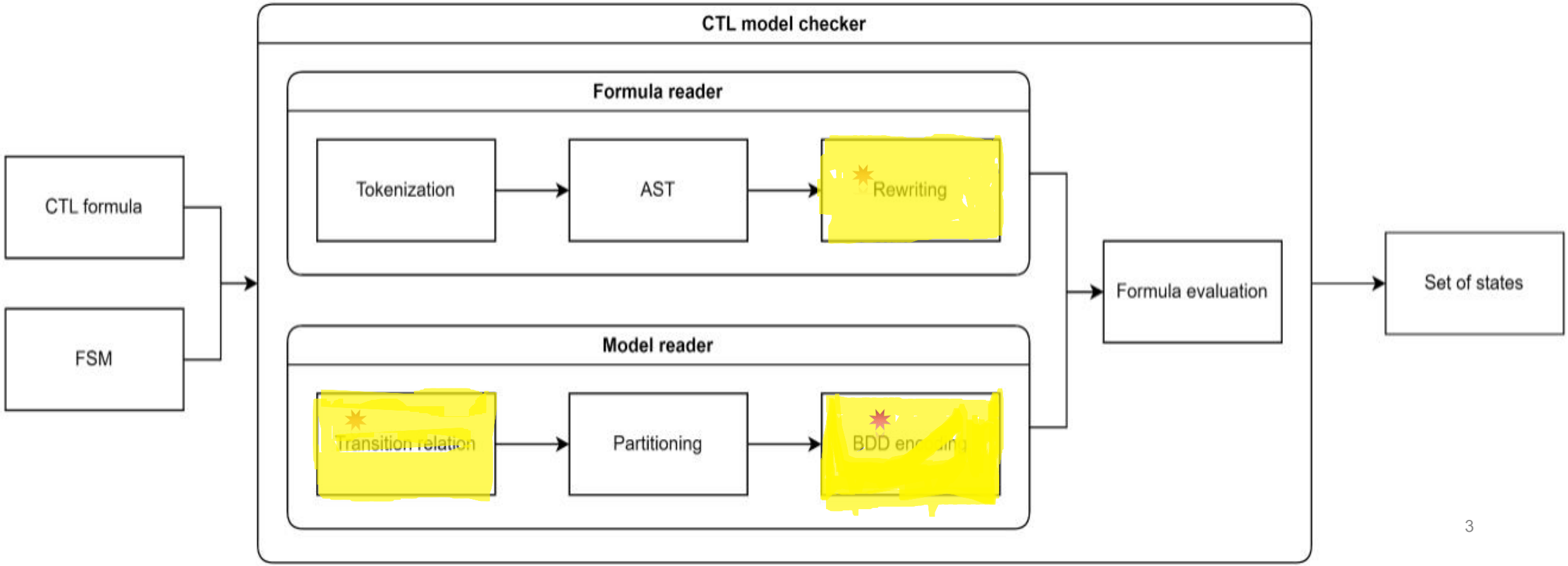


Goal

1. Work on BDDs and relation transition
2. Research Parallel Algorithms
3.Discover the best way to input FSM into our model checker

CTL model checking algorithm

--based on forward state traversal



Realization of Rewriting—— intruduction

1.double negations:

$$\neg(\neg f) = f$$

2.De Morgan's laws:

$$f \wedge g = \neg(\neg f \vee \neg g)$$
$$f \vee g = \neg(\neg f \wedge \neg g)$$

3.other logic equations:

$$EF\,f = E[true U\, f]$$
$$AX\,f = \neg EX \neg f$$
$$AF\,f = \neg EG \neg f$$
$$AG\,f = \neg EF \neg f$$
$$A[gU\,f] = \neg(E[\neg f U \neg g \wedge \neg f] \vee EG \neg f)$$

4. Replace path quantifiers:

$$s_0 \models f \iff p_0 \wedge f \neq \text{false}$$
$$s_0 \models f \iff p_0 \wedge \neg f = \text{false}$$

$$p \wedge \mathbf{E}[q\,\mathbf{U}\,f] \neq \text{false}$$
$$\iff \mathit{FwdUntil}\,(p, q) \wedge f \neq \text{false}$$

$$p \wedge \mathbf{EX}\,f \neq \text{false} \iff \mathit{Img}\,(p) \wedge f \neq \text{false}$$

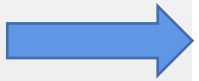
$$p \wedge \mathbf{EG}\,q \neq \text{false}$$
$$\iff \mathit{FwdGlobal}\,(p, q) \neq \text{false}$$

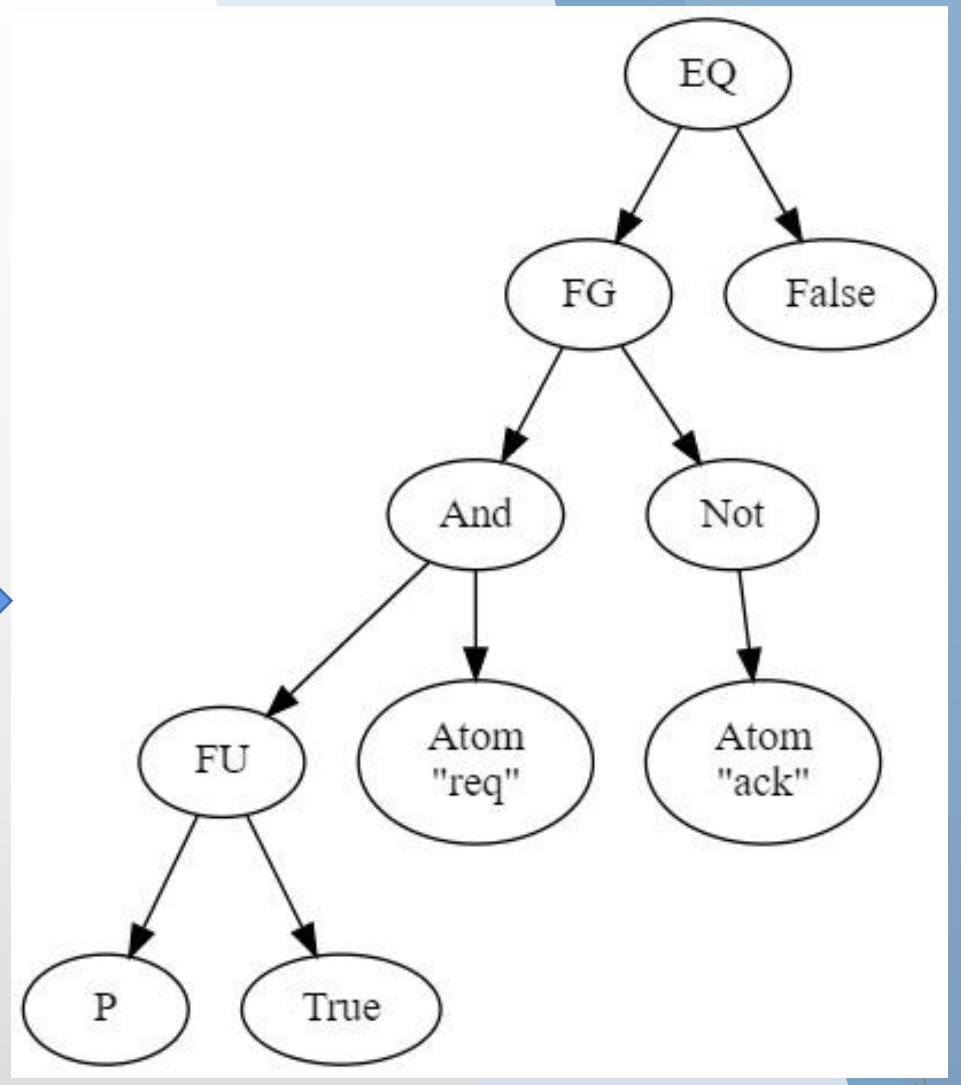
Realization of Rewriting

Simplify complex CTL formulas by using rewriting techniques

Example:

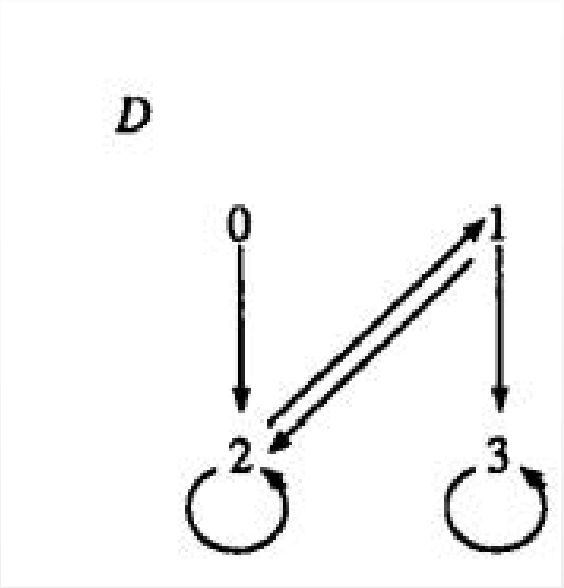
$$
\begin{aligned}
& s_0 \models \mathbf{AG}\,(req \rightarrow \mathbf{AF}\,ack) \\
\iff\ & s_0 \models \neg\mathbf{E}\,[\mathrm{true}\,\mathbf{U}\,(req \wedge \mathbf{EG}\,\neg ack)] \\
\iff\ & p_0 \wedge \mathbf{E}\,[\mathrm{true}\,\mathbf{U}\,(req \wedge \mathbf{EG}\,\neg ack)] = \mathrm{false} \\
\iff\ & \mathit{FwdUntil}\,(p_0, \mathrm{true}) \wedge (req \wedge \mathbf{EG}\,\neg ack) = \mathrm{false} \\
\iff\ & (\mathit{FwdUntil}\,(p_0, \mathrm{true}) \wedge req) \wedge \mathbf{EG}\,\neg ack = \mathrm{false} \\
\iff\ & \mathit{FwdGlobal}\,((\mathit{FwdUntil}\,(p_0, \mathrm{true}) \wedge req), \neg ack) = \mathrm{false}.
\end{aligned}
$$



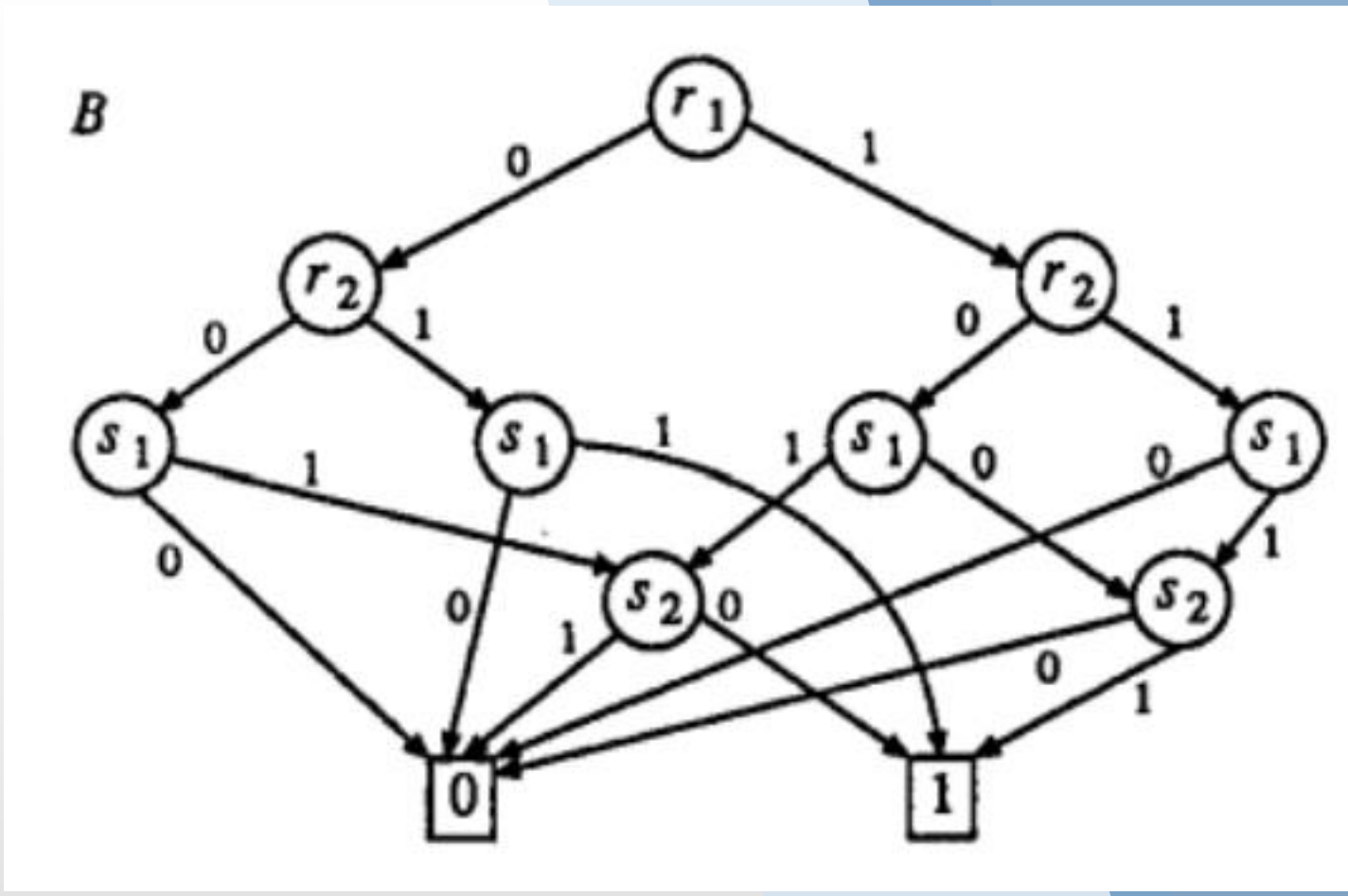


Produce the AST without temporal operators

Understanding of BDDs——Introduction



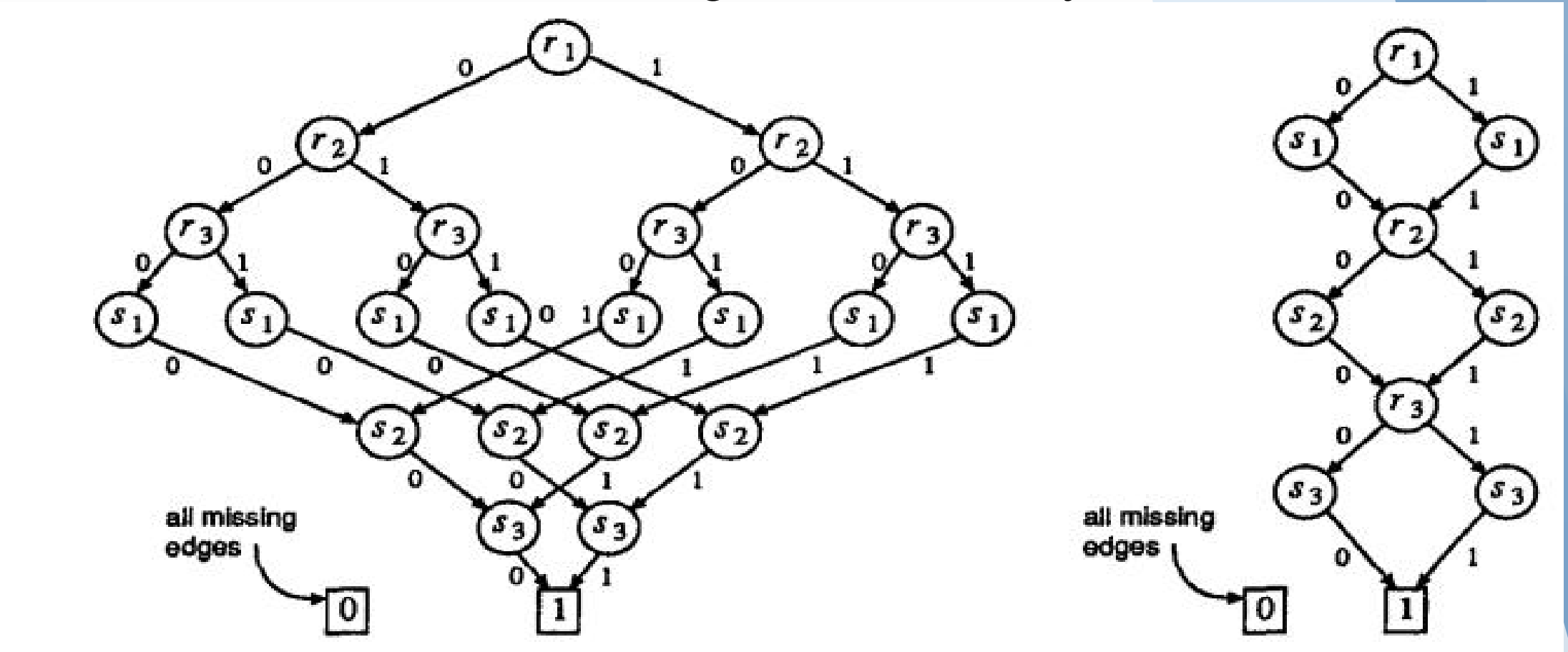
the relation D



BDD representation

Understandingof BDDs

——Encoding of transition systems as BDDs



BDD for (r1= s1) A (r2 = S2) A (r3 = S3) with poor (left) and good (right) variable ordering

Understandingof BDDs

——Encoding of transition systems as BDDs

State	Binary Encoding
0	000
1	001
2	010
3	011
4	100

Action	Binary Encoding
a	000
b	001
c	010
d	011
e	100
f	101
g	110

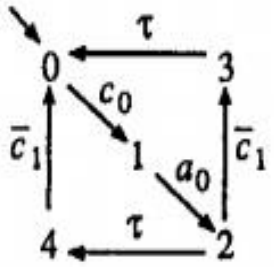
Transfer Relationship
(0,a,1)
(0,b,2)
(1,c,4)
(2,d,4)
(2,e,3)
(3,f,4)
(4,g,0)

We can express this transfer relation as a function X: B^9→B, where:

1. The first three digits correspond to the coding of the action
2. The middle three bits correspond to the encoding of the initial state
3. The last three digits correspond to the encoding of the target state

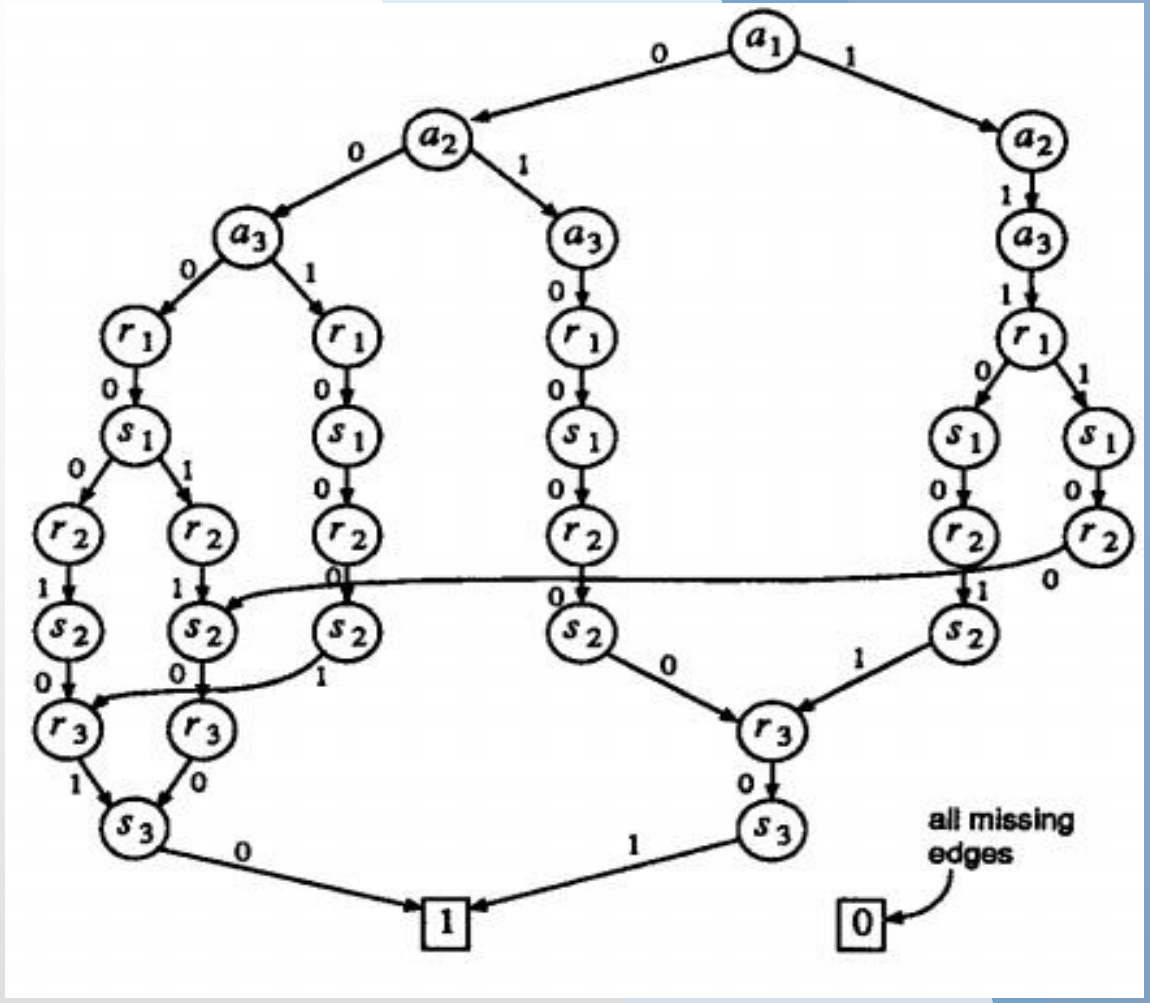
Understandingof BDDs

——Encoding of transition systems as BDDs



States			Actions		
0	↦	000	τ	↦	000
1	↦	001	a_0	↦	001
2	↦	010	$\bar{a}_0$	↦	101
3	↦	011	c_0	↦	010
4	↦	100	$\bar{c}_0$	↦	110
			c_1	↦	011
			$\bar{c}_1$	↦	111

A transition system and the encoding of its states and actions



BDD encoding of the transition relation of the system of left

Understanding of Partitioning transition relation

- Decompose a large state transition system into multiple small subsystems
- So that each subsystem can be processed separately
- Less time, more speed

States/Actions	A1	A2
S1	S2	S3
S2	S1	S1
S3	S1	S2

1. {s1, s2} as one part and {s3} as another part.

2. Individual CTL model checking for each sub-BDD, which can be done concurrently in parallel computing

What we want to do next step?

- **FIRST:**

Work on BDDs and transition relation;

Research Partitioning Algorithms

- **SECOND:**

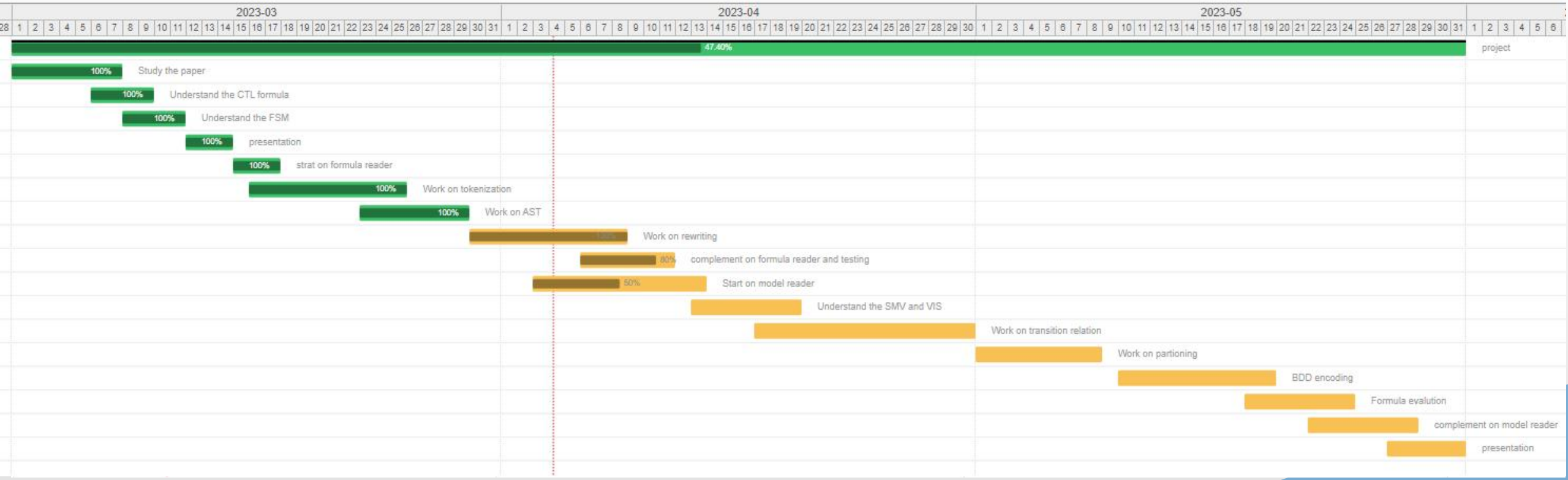
- **Discover the good way to input FSM into our model checker**

Goal—— Research diffenrent CTL models

Model	SMV	VIS	BINGO	
			Bwd	Fwd
gigamax	3.6sec 1.6MB	21.3sec 4.9MB	9.3sec 3.7MB	9.1sec 3.7MB
atm_sw	1734sec 8.3MB	*space*	9447sec 27.1MB	1158sec 23.1MB
dh_1	—	138.7sec 6.8MB	29.6sec 3.7MB	20.3sec 3.8MB
dh_2	—	*space*	*time*	1357sec 31.5MB
vpp	*time**	*space*	45569sec 17.5MB	223.3sec 13.2MB
pipe_s	23715sec 60.2MB	51.7sec 6.1MB	12429sec 5.5MB	64.6sec 4.9MB
pipe_d	—	*space*	*time*	8979sec 249.7MB

'*time*'	exceeded 24 hours in CPU time.
'*space*'	exceeded 300 megabytes of memory.
'*'	failed while building a transition relation.

Schedule


2023-03
2023-04
2023-05
project
Study the paper
Understand the CTL formula
Understand the FSM
presentation
strat on formula reader
Work on tokenization
Work on AST
Work on rewriting
complement on formula reader and testing
Start on model reader
Understand the SMV and VIS
Work on transition relation
Work on partioning
BDD encoding
Formula evalution
complement on model reader
presentation

Understandingof Partition transition relation

Leaf type	Function type
BDDs `false` and `true`	Total functions $\mathbb{B}^N \to \mathbb{B}$ (BDDs)
64-bit integer (`uint64`)	Partial functions $\mathbb{B}^N \to \mathbb{N}$
Floating-point (`double`)	Partial functions $\mathbb{B}^N \to \mathbb{R}$
Rational leaves	Partial functions $\mathbb{B}^N \to \mathbb{Q}$
GMP library leaves (`mpq`)	Partial functions $\mathbb{B}^N \to \mathbb{Q}$

Realization of Rewriting

not not p = p

Material conditional: $x \to y = \neg x \vee y$

Material biconditional: $x \leftrightarrow y = (x \wedge y) \vee (\neg x \wedge \neg y)$

De Morgan's laws:

$$x \wedge y = \neg(\neg x \vee \neg y)$$
$$x \vee y = \neg(\neg x \wedge \neg y)$$

$$\begin{aligned}
\mathbf{EF}\, f &= \mathbf{E}\,[\mathrm{true}\,\mathbf{U}\, f] \\
\mathbf{AX}\, f &= \neg\,\mathbf{EX}\,\neg f \\
\mathbf{AF}\, f &= \neg\,\mathbf{EG}\,\neg f \\
\mathbf{AG}\, f &= \neg\,\mathbf{EF}\,\neg f \\
\mathbf{A}\,[g\,\mathbf{U}\, f] &= \neg\,(\mathbf{E}\,[\neg f\,\mathbf{U}\,\neg g \wedge \neg f] \vee \mathbf{EG}\,\neg f)
\end{aligned}$$

$$s_0 \models f \iff p_0 \wedge \neg f = \mathrm{false}$$

$$p \wedge \mathbf{E}\,[q\,\mathbf{U}\, f] \neq \mathrm{false} \iff \mathit{FwdUntil}\,(p, q) \wedge f \neq \mathrm{false}$$

$$p \wedge \mathbf{EX}\, f \neq \mathrm{false} \iff \mathit{Img}\,(p) \wedge f \neq \mathrm{false}$$

$$p \wedge \mathbf{EG}\, q \neq \mathrm{false} \iff \mathit{FwdGlobal}\,(p, q) \neq \mathrm{false}$$

Understandingof transition relation

1. transition relation

states	actions	next states
off	open	on
on	close	off



2. boolean matrix

	on	off
on	0	1
off	0	1

BDD representation

- represent the state transition relation of the system.
- FSM model, it needs to be converted into a Boolean relation according to its state transition function.
- This Boolean relation can be represented as a relation matrix

THANK YOU